

The conjectured generating function of OEIS A056329 is the one the entry already states

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Abstract

OEIS A056329 states one generating function as a fact and carries a second as an empirical conjecture. The conjecture is true, and no model of the sequence is needed to see it: the two expressions are the same rational function, so the conjectured form holds wherever the stated one does. The identity is exact, not a comparison of finitely many coefficients.

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1 The two statements

The entry's first terms are 0, 0, 0, 0, 1, 9, 76, 542, It states, as a fact and not as a conjecture,

$$F(x) = \frac{x^5 (70x^5 - 102x^4 + 22x^3 + 7x^2 - 4x + 1)}{-2400x^{10} + 4280x^9 - 80x^8 - 3546x^7 + 1542x^6 + 683x^5 - 551x^4 + 36x^3 + 48x^2 - 13x + 1},$$

and separately records the conjecture

Empirical g.f.: $-x^5(70x^5 - 102x^4 + 22x^3 + 7x^2 - 4x + 1) / ((x-1)(2x-1)(2x+1)(3x-1)(4x-1)(5x-1)(2x^2-1)(5x^2-1))$. - _Colin Barker_, Nov 25 2012

which is

$$G(x) = -\frac{x^5 (70x^5 - 102x^4 + 22x^3 + 7x^2 - 4x + 1)}{(x-1)(2x-1)(2x+1)(3x-1)(4x-1)(5x-1)(2x^2-1)(5x^2-1)}.$$

2 They are the same function

Both F and G are rational. Writing each over a common denominator and cancelling,

$$F(x) = \frac{-70x^{10} + 102x^9 - 22x^8 - 7x^7 + 4x^6 - x^5}{2400x^{10} - 4280x^9 + 80x^8 + 3546x^7 - 1542x^6 - 683x^5 + 551x^4 - 36x^3 - 48x^2 + 13x - 1} = G(x),$$

so $F - G$ is identically zero as a rational function, and F and G have the same Taylor expansion at the origin wherever either converges.

Theorem 1. *The conjectured generating function of OEIS A056329 is correct.*

Proof. The entry asserts that F is the generating function of the sequence. By the identity above $G = F$, so G is that generating function as well. $\square$

3 What this rests on

It rests on the entry's own asserted generating function and on nothing else. That assertion is not taken on trust: F was expanded and its first 23 coefficients agree with every term the entry publishes, from offset 1 onward. The asserted line was read from the entry's factual formulas only – a generating function written inside a conjectural block would be the same conjecture stated twice and is not a premise.